

U22EC004



S.V. National Institute of Technology, Surat
Department of Electronics Engineering
END Semester Exam, DECEMBER 2024
B. Tech. (EC) 5th Semester
Digital Communication (EC 305)

Marks: 100

Time: 3 Hours

Instructions:

1. Answer all the questions in the sequential order.
2. Assume data wherever required.

		Marks	
Q1	ATTEMPT THE FOLLOWING QUESTIONS	[11]	CO 1
A.	Does a binary prefix-free code with codeword lengths 2,2,2,3,4 exists? Justify your answer.	[2]	
B.	An analog signal having 4kHz bandwidth is sampled at 1.25 times the nyquist rate and each sample is quantized into one of 256 equally likely levels. Assume that the successive samples are statistically independent i. What is the information rate of this source? ii. Can the output of this source be transmitted without error over an AWGN channel with a bandwidth of 10kHz and S/N ratio of 20dB? iii. Find the S/N ratio required for error-free transmission for part (i) iv. Find the bandwidth required for an AWGN channel for error-free transmission of the output of this source if the S/N ratio is 20dB?	[4]	
C.	Explain the basic principle of Orthogonal frequency Division multiplexing (OFDM). Sketch the basic blocks of implementation of OFDM.	[5]	
Q2	ATTEMPT THE FOLLOWING QUESTIONS	[16]	CO 2
A.	Derive the power spectrum density (PSD) of binary signal using differential code (See Figure 1.) with half width rectangular pulses. Determine the PSD $S_y(\omega)$.	[6]	
	Figure 1.		
B.	For a BPSK modulator with a carrier frequency of 70MHz and an input bit rate of 10Mbps, determine the minimum Nyquist bandwidth, and also calculate the maximum and minimum upper and lower side frequencies.	[4]	
C.	Explain the significance of eye diagram tool in digital communication. Also by sketching the eye diagram for ideal pulses define maximum opening point, sensitivity to timing jitter and level crossing jitter.	[6]	
Q3	ANSWER THE FOLLOWING QUESTIONS	[19]	CO 3
A.	An audio signal of bandwidth 4kHz is sampled at a rate 25% over nyquist rate and quantized. The quantization error is not to exceed 0.1% of the signal peak amplitude. The resulting quantized samples are now coded and transmitted by 4-ary pulses. i. Determine the minimum number of 4-ary pulses required to encode each sample. ii. Determine the minimum transmission bandwidth required to transmit this data with zero ISI. iii. If 4-ary pulses satisfy the nyquist criterion with 25% roll-off factor used to transmit this data, determine the transmission bandwidth.	[4]	
B.	A source is producing sequence of independent symbols A, B, C and D with probabilities $p(A)=0.5$, $p(B)=0.25$, $p(C)=0.125$ and $p(D)=0.125$ i. Devise an unambiguous Shannon-Fano code for the output of this source. ii. Compute the coding efficiency of your code.	[4]	

C. In a binary source, 0s occur three times as often as 1s. Find the information contained in the 0s and 1s. Express the information in bits, nats and Hartleys. Also find the entropy of the binary source. If the output data rate of the source is 16000 symbols/second, find the entropy of the source in bits/second. [5]

D. The state diagram for (2,1,2) convolution encoder is shown in below Figure 2. Determine the output sequence and the sequence of states which the encoder goes through using trellis code when an input sequence of 1 1 0 1 1 0 0 1 0 is given. Assume an initial state of 0 0 i.e. 'a'. [6]

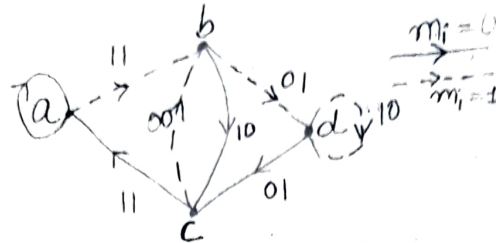


Figure 2.

Q4. ATTEMPT THE FOLLOWING QUESTIONS [17] CO 4

A. In a binary data transmission using duobinary pulses, samples values were read as follows:

1 2 0 0 0 -2 0 0 -2 0 2 0 0 -2 0 2 2 0 -2

- Explain if there is any error in the detection.
- Can you guess the correct transmitted digit sequence? If there is more than one possible correct sequence, give as many as possible correct sequences assuming that more than one detection error is extremely unlikely.

B. The bitstream 1 1 0 1 1 1 0 0 1 0 1 is to be transmitted using differential phase shift keying (DPSK). Determine the encoded sequence and the transmitted phase sequence. Also, find the demodulated signal. [6]

C. A scrambler is shown in the Figure 3 below. Design the corresponding descrambler. If a sequence $S = 101010100000111$ is applied to the input of this scrambler, determine the output sequence T . verify that if this T is applied to the input of descrambler, the output is the descrambler S . [6]

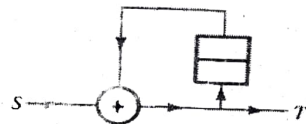


Figure 3.

Q5. ATTEMPT THE FOLLOWING QUESTIONS [17] CO 5

A. For the binary signaling with received peak amplitude $A_p = 0.0015$ determine the detection error probability if the channel noise is Gaussian with rms value 0.0003.

Given: $Q(5) = 2.867 \times 10^{-5}$; $Q(2.5) = 0.00621$

- Assuming zero ISI, what is the error probability of polar signalling, ON-OFF signalling and bipolar signalling? [3]
- Determine the received power in each of the three cases assuming half-width rectangular pulses. [3]
- In order to achieve $P(\epsilon)$ similar to that in the polar case, what must be the received power for the ON-OFF and bipolar cases? [2]

B. Sketch the QPSK waveform for the sequence 1 1 0 1 0 1 0 0 1 0, assuming the carrier frequency to be equal to bit rate. [4]

C. Data stream encoded using a (3,1) repetition code, is transmitted over a binary symmetric channel having a transition probability of $P=10^{-2}$. Determine the probability of error P_e of the symbols received at the destination. [5]

Q6

ATTEMPT THE FOLLOWING QUESTIONS

[20]

CO 6

A.

Figure 4 below shows a binary data transmission scheme. The baseband signal generator uses full-width pulses and polar signaling. The data rate is 1 Mbits/sec.

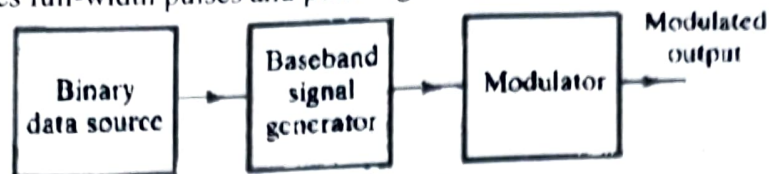


Figure 4

- If the modulator generates a PSK signal, what is the bandwidth of the modulated output?
- If the modulator generates FSK with difference $f_{c1} - f_{c2} = 100\text{kHz}$, determine the modulated signal bandwidth.
- If instead of full-width pulses, Nyquist criterion pulses with roll-off factor $r=0.2$ are used, then what's the bandwidth of the modulated output?
- If multi-amplitude scheme with $M=4$ (PAM signal with full-width pulse is used), then determine the bandwidth of the modulated output?

B.

A systematic (7,4) cyclic code is to be generated making use of $x^3 + x + 1$ as the generator polynomial. Draw the block schematic diagram of the encoder. By tracing the contents of the shift registers after shifting-in each of the message bits, determine the parity check bits and the codeword in systematic form, assuming a message sequence $m = \{1111\}$.

C.

A four stage shift register with feedback connections taken from the outputs of stages 4 and 1 through a modulo-2 adder, is used for PN sequence generation. Assuming the initial contents of the shift register to be 0100, determine the output sequence. What is the length of the sequence?

D.

Binary data is transmitted using a pulse $p(t)$ for '0' and a pulse $3p(t)$ for '1'. Show that the optimum receiver for this case is a filter matched to $p(t)$ with the detection threshold $2E_p$ as shown in the Figure 5. Determine the error probability P_b of this receiver as a function of E_b/N if 0 and 1 are equiprobable.

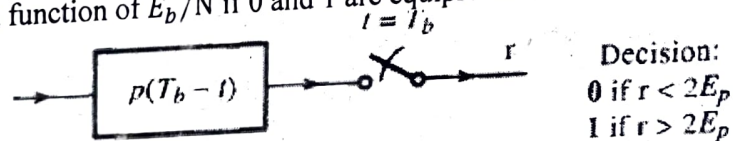


Figure 5.

****ALL THE BEST****